DNAStoreAI

Optimizing Digital Data Storage with Machine Learning

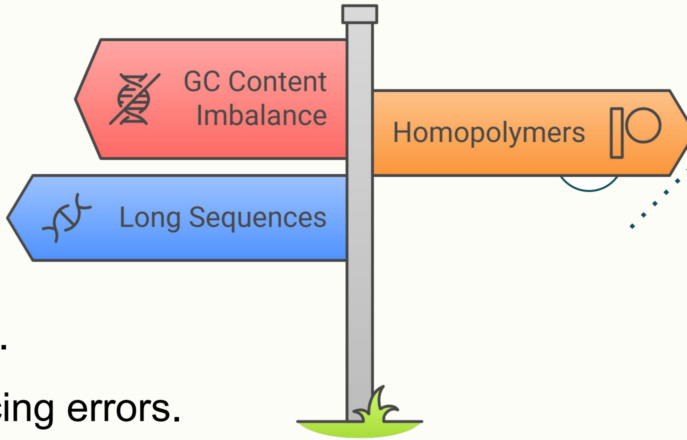
Encoding Digital Data into Stable DNA Sequences

Problem Statement

“Digital data is exploding. Storage media? Not evolving fast enough.”

- Traditional storage (HDDs, SSDs) degrades over time.
- DNA offers ultra-dense and durable storage, but...
- **Challenges:**
  - Imbalanced **GC content** causes structural issues.
  - **Homopolymers** (e.g., AAAAA...) lead to sequencing errors.
  - Long sequences without constraints result in high failure rates.

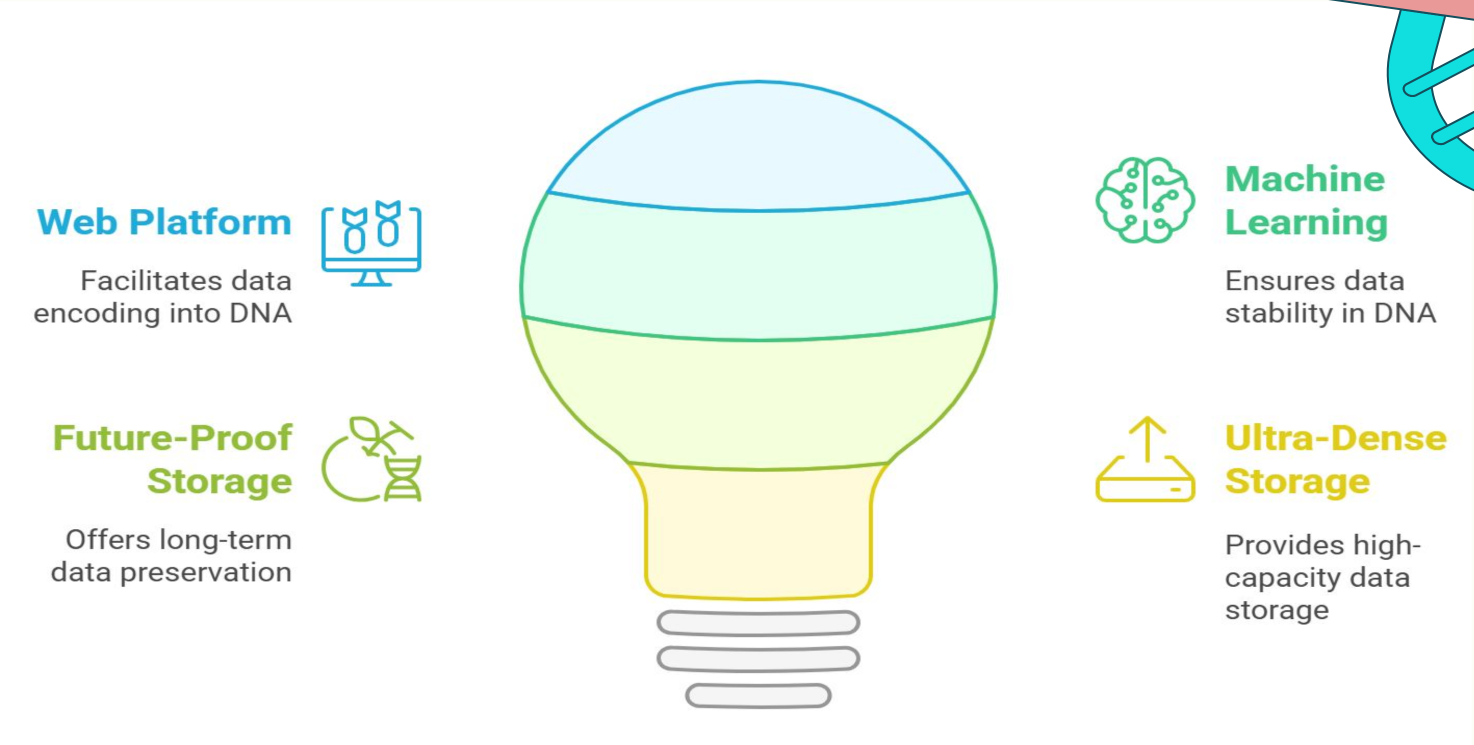
Challenges in DNA data storage



Uniqueness and Strength

- First-of-its-kind **interactive web platform**.
- Real-time **biological safety feedback**.
- Extensible to **full genome encoding**, bioart, cryptoDNA.
- Fully documented and deployable.

What is DNAStoreAI?

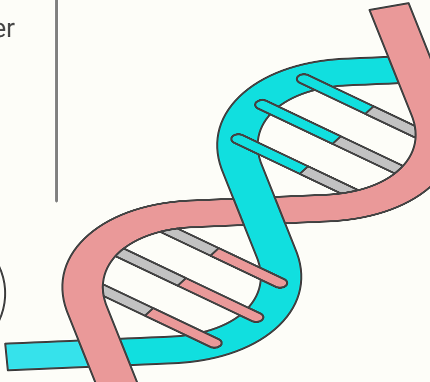


Why Machine Learning ?

"Rules alone can't adapt to complex sequence interactions."

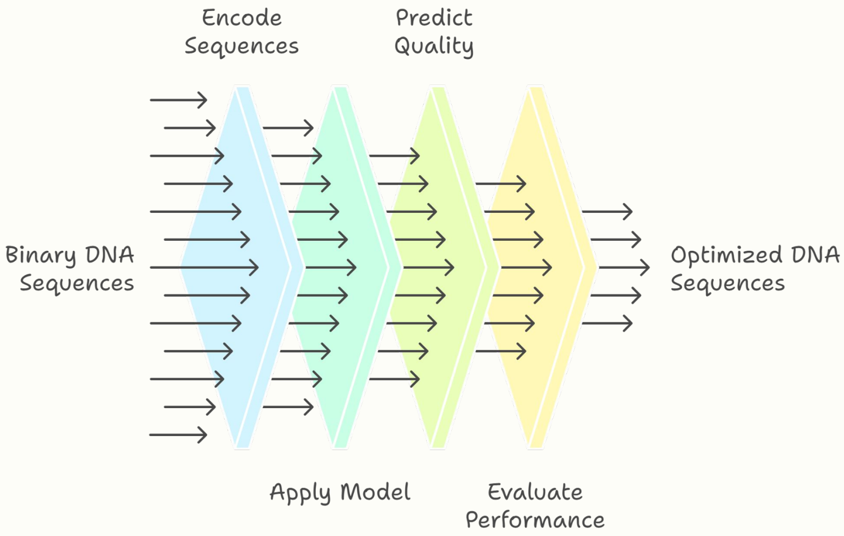
- DNA design has nonlinear constraints.
- ML can **learn from patterns** that heuristics can't catch.
- Our model helps:
  - Predict unstable regions.
  - Optimize encoding for GC/homopolymer balance.
  - Learn from both real and simulated datasets.

Pros	VS	Cons
Pattern learning		Computational cost
Nonlinear constraints		Data dependency
Unstable region prediction		Interpretability challenges
GC/homopolymer optimization		
Real and simulated data		



ML Model Architecture

- **Input:** Binary-to-encoded DNA sequences.
- **Output:** Quality predictions or optimized versions.
- **Model Type:** (e.g., CNN, RNN, Transformer, or Heuristic with scoring).
- **Dataset:** Real and simulated examples of stable/unstable DNA.
- **Visuals:**
  - Model architecture diagram.
  - Training curves (loss, accuracy).



Visual Comparison

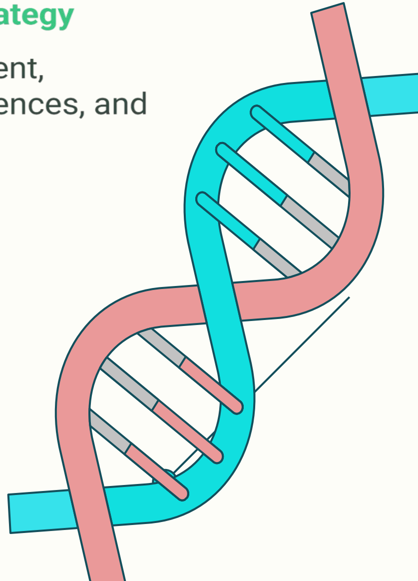
Metric	Naive	ML-Optimized
GC Content	80%	✅ 49–51%
Homopolymers	6bp	✅ ≤3bp
Sequence Length	Long	✅ Compressed
Error Risk	High	✅ Low

Naive Strategy

High GC content, long sequences, and high error risk

ML-Optimized Strategy

Balanced GC content, compressed sequences, and low error risk



Tech Stack

Frontend	React + Tailwind + Framer
Backend	Flask
Ml Model	Tensorflow.js
Visuals	Chat.js